Param Nikhil Shah

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Education

Sardar Patel Institute of Technology (S.P.I.T.) ***Aug'23 – July'27(Exp)***
Bachelor of Technology in Computer Engineering *Mumbai*
Minor in Internet of Things *CGPA: 8.26*

Experience

Barclays | Technology Summer Intern ***Jun'26 – Aug'26***
Java, Spring Boot, REST APIs, Thymeleaf, JavaScript, CSS, Maven, Linux, PuTTY, WinSCP *Pune, India*

- Built an internal application-health dashboard using **Spring Boot**, **REST APIs**, **Thymeleaf** and **JavaScript** to monitor application status across multiple environments.
- Implemented **automated recovery** and **one-click start/stop/repair** workflows for failed applications, reducing manual recovery steps for support users.

Projects

FOLIUM: Explainable Causal Intelligence for Behavioural Bias in Investing ***Feb'26 – Oct'26***
Python, FastAPI, PostgreSQL, Redis, XGBoost, SHAP, DoWhy, NumPy, pandas, React, TypeScript

- Developed the full-stack platform with **FastAPI, PostgreSQL (Row-Level Security), Redis, and React + TypeScript**, with **JWT/OAuth2** auth, live **SSE** progress, and **630+ tests**.
- Engineered a **causal, decision-time behavioral bias detection engine** using **XGBoost** (**87.9% accuracy**, 0.84 Macro F1) with **per-prediction SHAP attributions**, and **DoWhy** causal inference to quantify how much each bias drives a sell decision.
- Built a **personalized Monte Carlo simulator** (**NumPy/pandas**, real NIFTY and gold history) projecting **HOLD vs SELL after-tax** wealth over up to 30 years, with P10/P50/P90 uncertainty bands.
- Designed an **Explainable AI layer** with bilingual (English/Hindi) **Jinja2** explanations and a **SHA-256 hash-chained audit trail** that makes every analysis verifiable and tamper-evident.

ARCHON: AI Engineering Intelligence Platform ***Sep'26 – Oct'26***
Python, FastAPI, tree-sitter, PostgreSQL, SQLite, Next.js, TypeScript, Gemini API, Groq API, Docker

- Built a **FastAPI** backend using **tree-sitter** to parse **6 languages** (Python, JS/TS, Go, Java, SQL, Prisma) into a **cross-file code graph** of imports, calls and inheritance, stored in **SQLite/PostgreSQL**.
- Implemented **codebase chat** with hybrid **BM25 + Gemini embedding** retrieval and cited answers, backed by a provider-agnostic **LLM layer** with automatic **Gemini/Groq** fallback to handle free-tier rate limits.
- Developed **security, performance and tech-debt analyzers** on a shared **Finding** model, plus **13 AI tools** for root-cause debugging, change-impact scoring, test generation and diff code review.
- Created an interactive **Next.js + TypeScript** code map using **Canvas2D, d3-zoom and Web Workers** for smooth rendering of large graphs; containerized with **Docker Compose** and covered by a **35-test pytest** suite.

CareerPath360.AI ***Aug'25 – Dec'25***
React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Python, FastAPI, NLP, REST APIs

- Built the **Node.js/Express.js** backend with **JWT authentication**, **REST API** routes, and **MongoDB** persistence for the resume builder, and **deployed** the full-stack app on **Render**.
- Developed a **Python-based ML microservice** using **FastAPI** for resume parsing, ATS scoring, and job matching, applying **NLP** with **Sentence Transformers (BERT)** and **feature engineering** for skill extraction and resume–JD semantic similarity.
- Trained an **XGBoost** model within a **hybrid ML + rule-based scoring pipeline** with **robust fallbacks** for realistic ATS scores, using **Gemini** for grounded, LLM-based resume feedback.

Skills

Languages: Java, Python, TypeScript, JavaScript, C, SQL, HTML, CSS
Frameworks & Libraries: Spring Boot, Thymeleaf, React.js, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS, Jinja2, NumPy, Pandas
AI/ML: XGBoost, DoWhy, SHAP, Sentence Transformers (BERT), Monte Carlo Simulation, AutoGen
Databases & Tools: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Git, GitHub, Maven, pytest, tree-sitter
Cloud & DevOps: Render, Vercel, Docker, Docker Compose, CI/CD, Linux, PuTTY, WinSCP
Core Concepts: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Database Management Systems, Machine Learning, RESTful APIs, JWT/OAuth2, Agile Development

Positions of Responsibility

S.P.I.T. Placement Office | Training & Placement Coordinator ***Dec'23 – Present***

- Served as the primary point of contact between recruiters and students, managing campus drives and pre-placement sessions and analyzing placement data to identify target companies.

S.P.I.T. Sports Committee | Head of Marketing and Tournament Admin ***Sep'23 – Aug'25***

- Led marketing, PR, and event operations for large-scale sports events, including a Half Marathon (1,500+ participants) and Spoorthi Sports Fest (5,000+ participants), ensuring seamless execution and strong brand visibility.